SOFTWARE REQUIREMENTS SPECIFICATION

FOR

SKILL EXCHANGE WEBSITE

A Software Requirements Specification Document

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1. Introduction

1.1 Purpose

The purpose of the Skill Exchange Website is to provide an online platform where users can share their skills and learn skills from other users. The system allows users to create profiles, list the skills they can offer, search for required skills, and send or receive skill exchange requests. The system also provides administrative functions for managing users and platform data.

1.2 Scope

The Skill Exchange Website connects people who want to teach a skill with people who want to learn that skill. Users can register and log in, maintain their profiles, add skills, search for other users, request an exchange, and manage exchange activities. The platform may also support payment processing when a paid service is selected. An administrator manages users, skills, requests, and other system information.

1.3 Objectives

- Provide a simple platform for users to exchange knowledge and skills.
- Allow users to create and manage their skill profiles.
- Help users find people who offer the skills they want to learn.
- Provide a structured process for sending and managing skill exchange requests.
- Provide secure payment processing when required.
- Allow administrators to monitor and manage the website.

1.4 Definitions and Abbreviations

SRS: Software Requirements Specification

User: A registered person who uses the website.

Admin: An authorized person who manages the website.

Skill: A knowledge area or ability offered or requested by a user.

Skill Exchange: An arrangement in which users share or learn skills from one another.

Payment Gateway: A service used to process online payments securely.

2. Overall Description

2.1 Product Perspective

The Skill Exchange Website is a web-based application. It acts as an intermediate platform between users who offer skills and users who want to learn them. The system consists of a user interface, application logic, database, and optional payment gateway integration.

2.2 Product Functions

- User registration and login.
- Profile creation and profile updates.
- Adding, editing, and removing offered skills.
- Searching and browsing available skills.
- Viewing other users and their skill information.
- Sending, accepting, or rejecting skill exchange requests.
- Managing exchange status.
- Processing payments when applicable.
- Admin management of users, skills, and platform activities.

2.3 User Classes and Characteristics

User: Can register, log in, manage a profile, offer skills, search for skills, and manage exchange requests.

Admin: Can access administrative functions and manage users, skills, requests, and system information.

2.4 Operating Environment

- Web browsers such as Google Chrome, Microsoft Edge, or Mozilla Firefox.
- Internet connection.
- Web/application server.
- Database server for storing user and transaction information.
- Payment gateway service for online payment processing, if enabled.

2.5 Design Constraints

- The website requires an internet connection.
- Only registered users can access user-specific functions.
- Administrative functions require authorized admin access.
- Payment processing depends on the selected payment gateway.
- User data must be stored and accessed securely.

2.6 Assumptions and Dependencies

It is assumed that users provide valid information during registration and have access to a compatible web browser. The availability of payment services depends on the selected payment gateway. The system also depends on the availability of the web server, database, and network.

3. Functional Requirements

FR-01 – User Registration: The system shall allow a new user to create an account using required personal and login information.

FR-02 – User Login: The system shall authenticate registered users before providing access to protected features.

FR-03 – Profile Management: The system shall allow users to create, view, and update their profiles.

FR-04 – Skill Management: The system shall allow users to add, edit, and remove skills they can offer.

FR-05 – Skill Search: The system shall allow users to search and browse available skills.

FR-06 – User Search: The system shall allow users to view suitable users based on their skills.

FR-07 – Exchange Request: The system shall allow a user to send a skill exchange request to another user.

FR-08 – Request Management: The system shall allow users to accept or reject received exchange requests.

FR-09 – Exchange Status: The system shall display the current status of an exchange request.

FR-10 – Payment: The system shall support payment processing when a paid skill/service is selected.

FR-11 – Payment Confirmation: The system shall display or store the result of a completed payment.

FR-12 – Admin Login: The system shall provide secure login for authorized administrators.

FR-13 – Admin Management: The administrator shall be able to manage users, skills, requests, and relevant system data.

FR-14 – Data Management: The system shall store and retrieve required information from the database.

4. Non-Functional Requirements

Performance: The system should respond to normal user actions within a reasonable time.

Security: User credentials and personal information should be protected from unauthorized access.

Usability: The interface should be simple, clear, and easy to navigate.

Reliability: The system should operate consistently and handle errors without losing important data.

Availability: The website should be available whenever the server and network services are operational.

Scalability: The system should be capable of supporting additional users and skills as usage grows.

Maintainability: The system should be organized so that future changes and updates can be implemented easily.

5. External Interface Requirements

User Interface: The website should provide registration, login, profile, skill search, exchange request, payment, and admin screens.

Software Interface: The application may communicate with a database and external payment gateway through appropriate interfaces/APIs.

Database Interface: The application shall store user, skill, exchange, and payment-related data in a database.

Payment Interface: If enabled, the system shall communicate with a payment gateway for payment authorization and confirmation.

Communication Interface: The website shall use standard web communication protocols such as HTTPS where supported.

6. System Features

6.1 User Registration

- The user opens the registration page.
- The user enters required details.
- The system validates the information.
- If valid, the system creates the user account.
- The system displays registration confirmation.

6.2 User Login

- The user enters login credentials.
- The system validates the credentials.
- If valid, the user is redirected to the appropriate dashboard.
- If invalid, the system displays an error message.

6.3 Profile and Skill Management

- The user opens the profile section.
- The user adds or updates profile information.
- The user adds skills that they can offer.
- The system stores the updated information.

6.4 Skill Search and Exchange

- The user searches for a required skill.
- The system displays matching skills/users.
- The user selects a suitable profile.
- The user sends an exchange request.
- The receiving user accepts or rejects the request.
- The system updates the request status.

6.5 Payment

- The user selects a paid service if available.
- The system displays the payment amount.
- The user proceeds to the payment gateway.
- The payment gateway processes the transaction.
- The system records the payment result.

6.6 Admin Management

- The administrator logs in.
- The system verifies admin credentials.
- The admin views and manages users and platform information.
- The admin can monitor relevant activities and maintain system data.

7. Use Case Requirements

The main actors in the system are User and Admin. The Payment Gateway is an external system involved in payment processing.

7.1 Main Use Cases

- Register Account
- Login
- Manage Profile
- Manage Skills
- Search Skills
- View User Profile
- Send Exchange Request
- Accept/Reject Exchange Request
- Make Payment
- View Payment Status
- Admin Login
- Manage Users
- Manage Skills and Requests

7.2 Use Case Description – Send Skill Exchange Request

Actor: User

Precondition: The user is registered and logged in.

Main Flow: User searches for a skill, selects a suitable profile, and sends an exchange request.

Alternative Flow: If the selected user is unavailable or the request cannot be submitted, the system displays an appropriate message.

Postcondition: The request is stored with a pending status and can be managed by the receiving user.

7.3 Use Case Description – Make Payment

Actor: User

External Actor: Payment Gateway

Precondition: The user has selected a paid service.

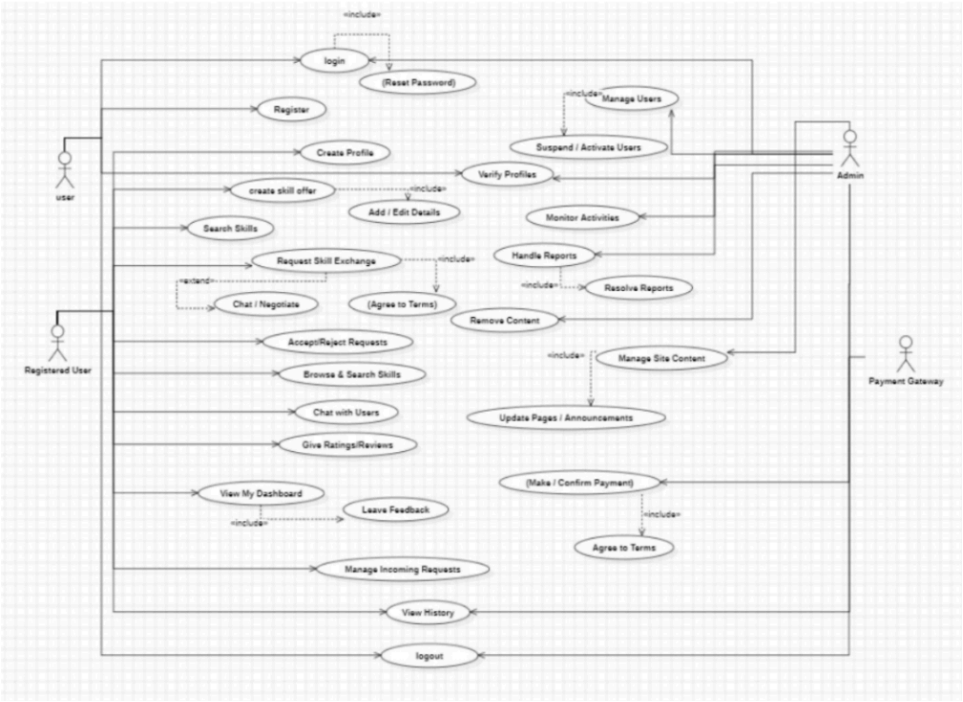
Main Flow: The system displays the amount, redirects the user to the payment gateway, and receives the payment result.

Postcondition: The system stores the payment status and displays confirmation or failure information.

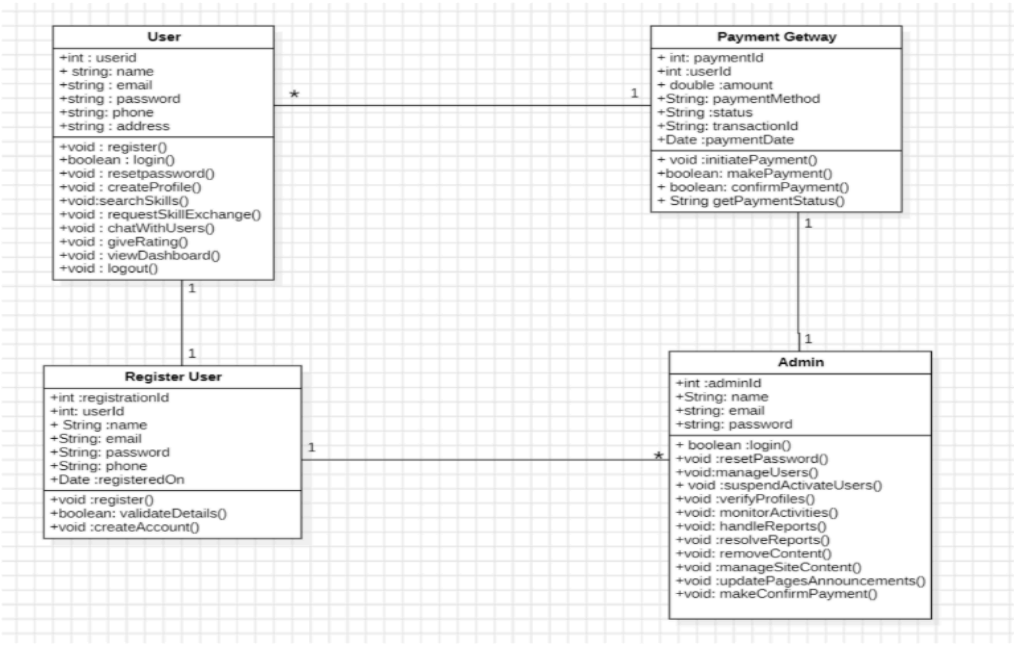
8. UML and System Models

The following UML diagrams can be included in the final SRS to represent the structure and behavior of the Skill Exchange Website:

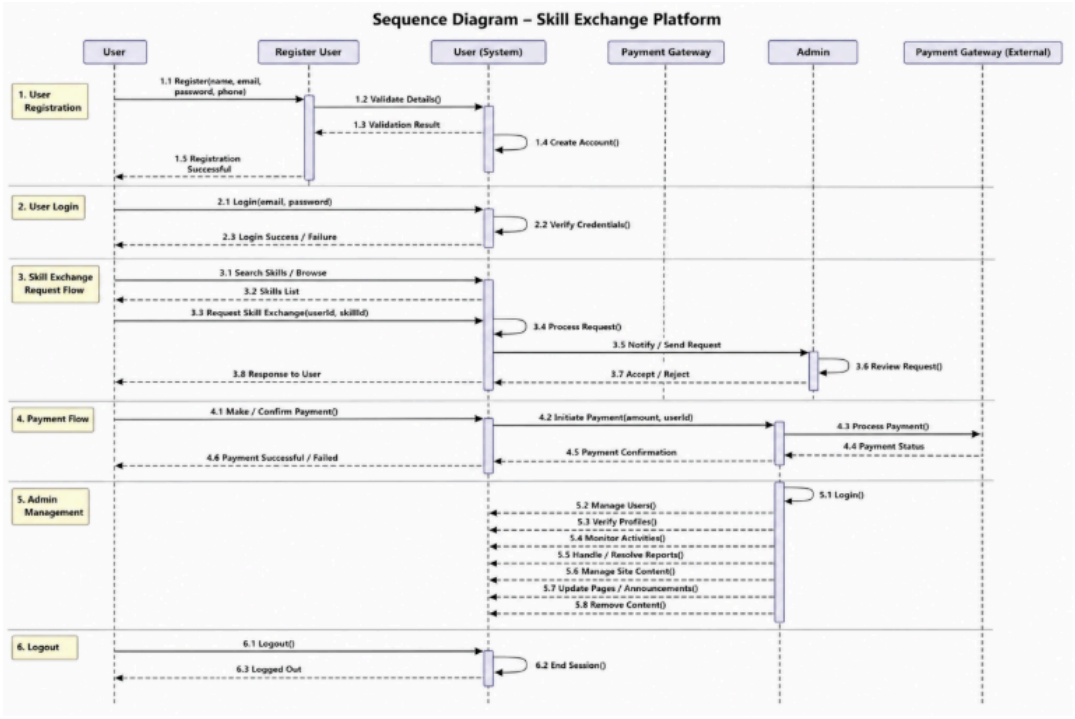
- Use Case Diagram – shows interactions between User, Admin, and Payment Gateway.



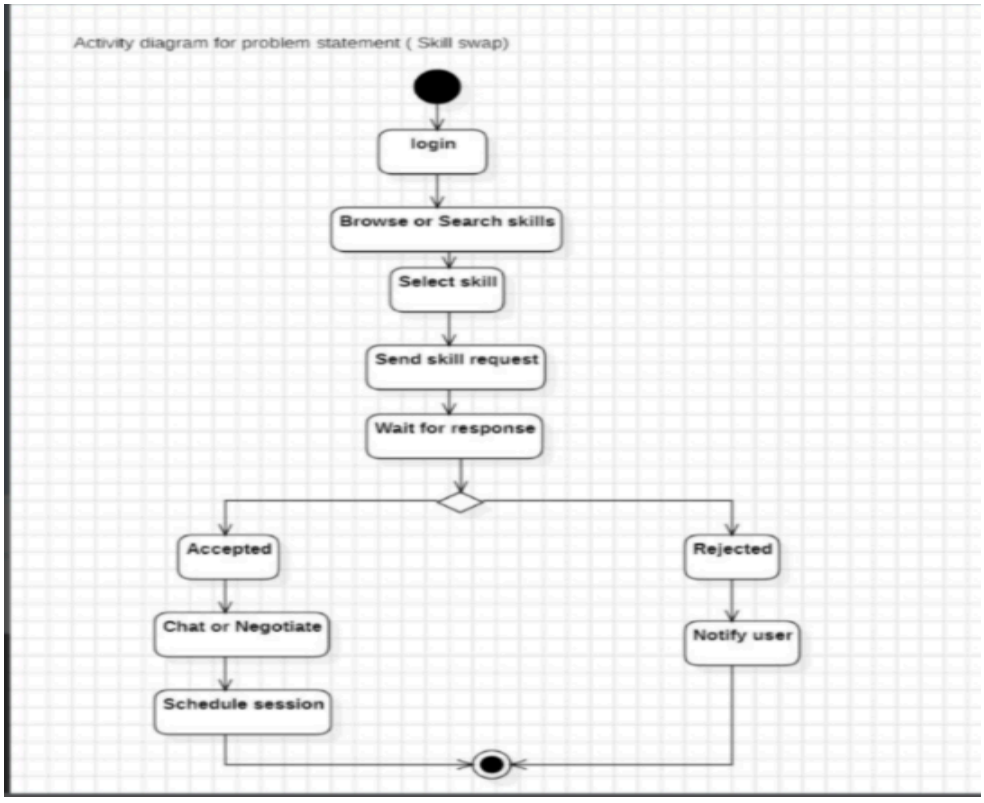
- Class Diagram – shows classes such as User, Register User, Payment Gateway, Admin, and related entities.



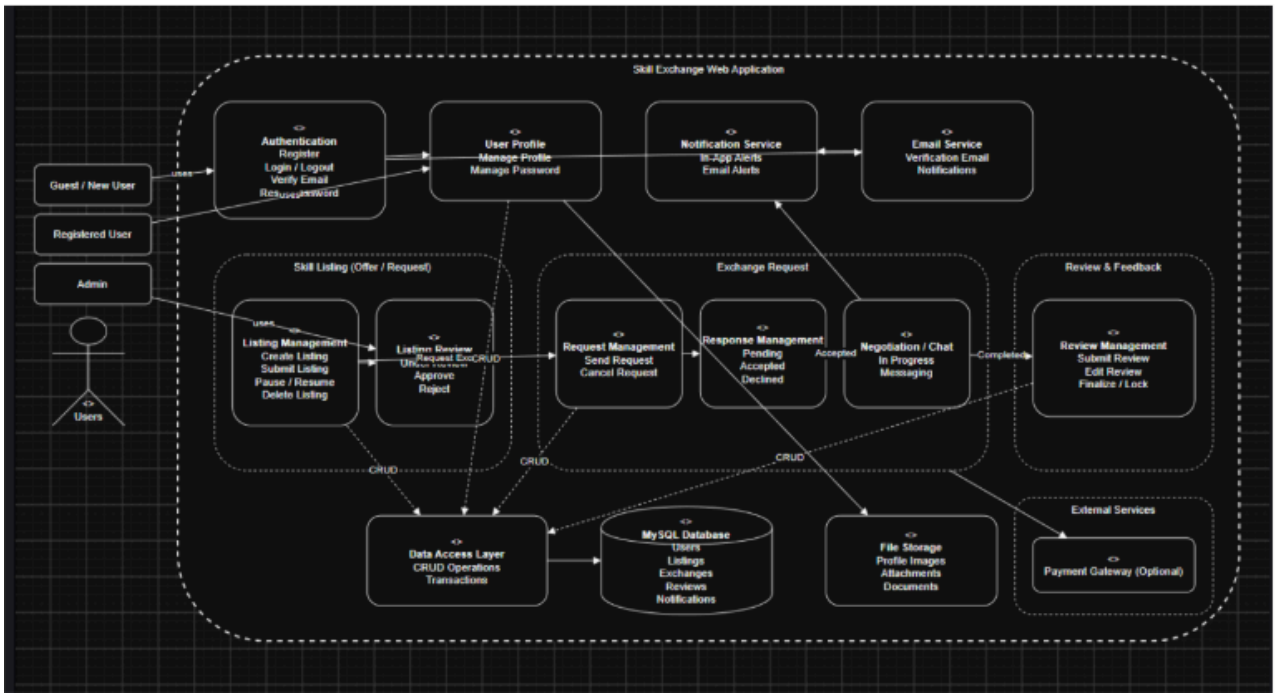
- Sequence Diagram – shows the sequence of interactions for important operations such as registration, login, skill exchange, and payment.



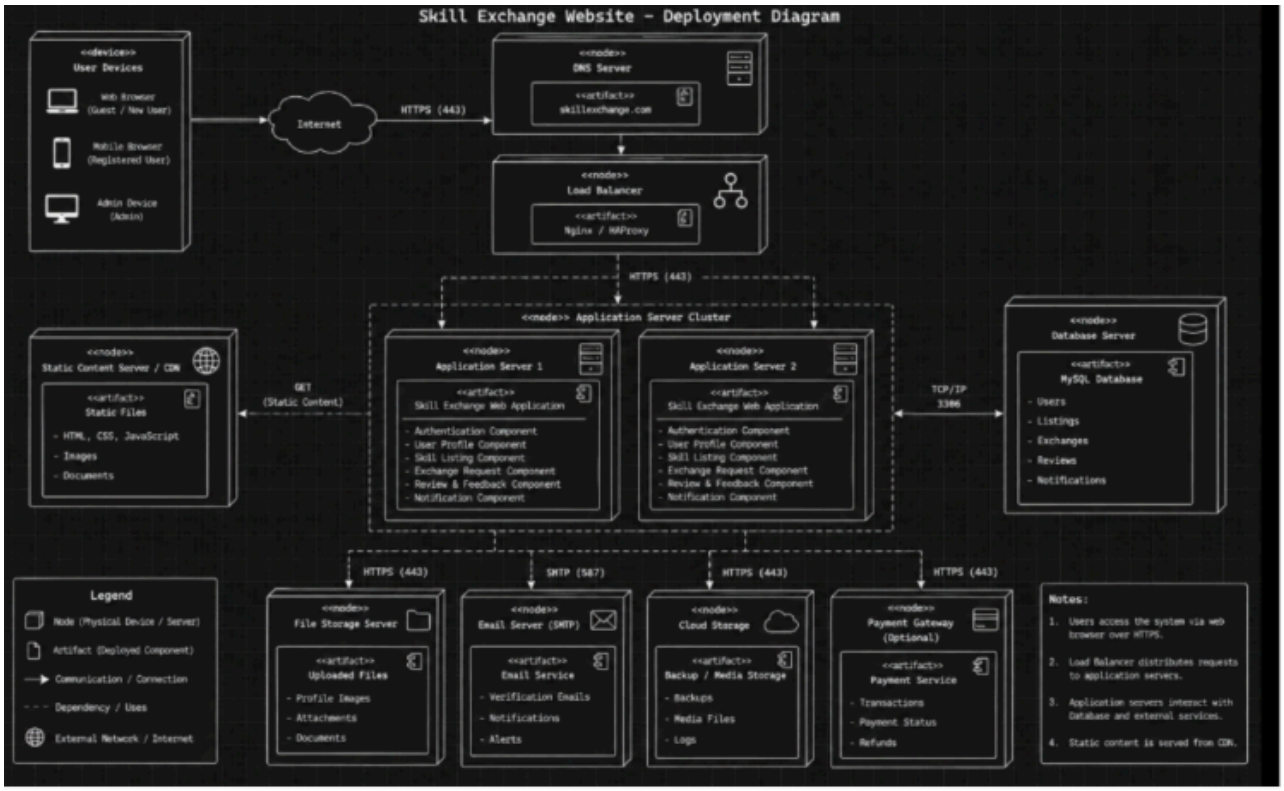
- Activity Diagram – shows the workflow of activities in the system.



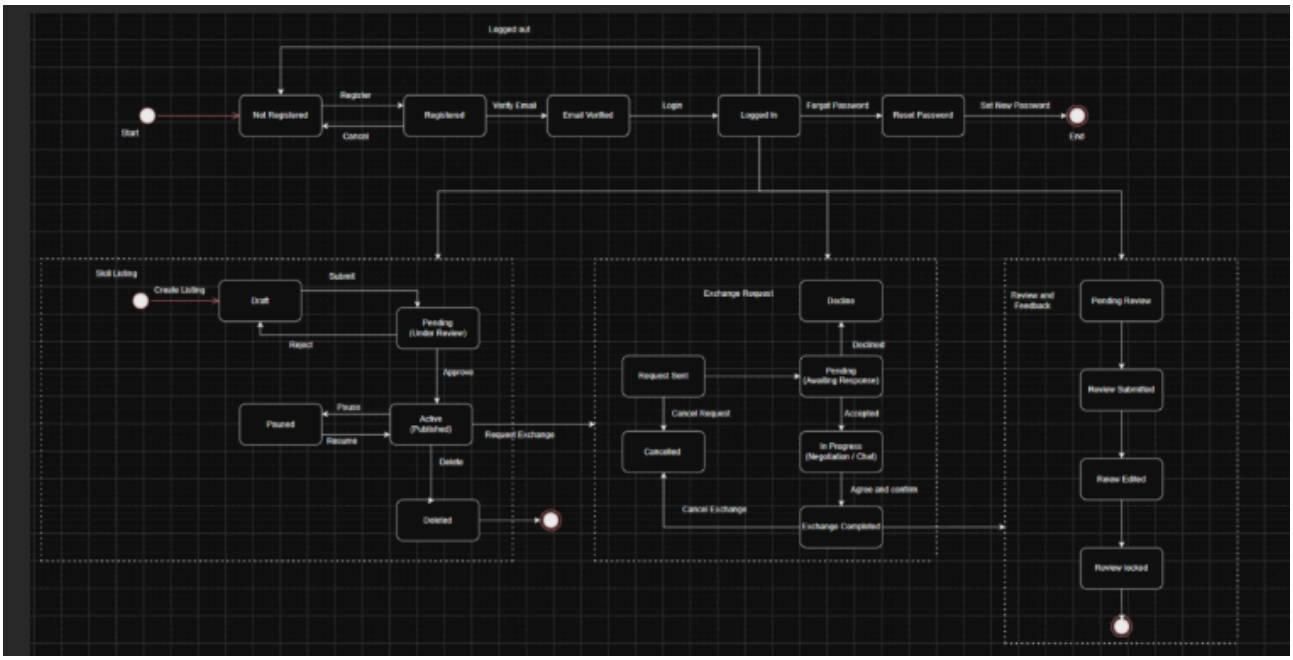
- Component Diagram – shows major software components and their dependencies.



- Deployment Diagram – shows how the web application, server, database, and external services are deployed.



- state diagram - A State Diagram shows how an object or system changes from one state to another when an event/action happens.



9. Database Requirements

The database should store information required for users, skills, exchanges, and payments. A possible logical structure is:

User: user_id, name, email, phone, password, profile_details

Skill: skill_id, skill_name, description, user_id

Exchange Request: request_id, sender_id, receiver_id, skill_id, status, request_date

Payment: payment_id, user_id, request_id, amount, payment_status, payment_date

Admin: admin_id, name, email, password

10. Security Requirements

- Only authenticated users should access protected user functions.
- Administrator functions should be accessible only to authorized administrators.
- Passwords should not be stored or transmitted in plain text.
- Sensitive communication should use secure protocols such as HTTPS.
- Payment information should be handled through the selected payment gateway and not unnecessarily stored by the application.
- Input validation should be used to reduce invalid or malicious data.

11. Assumptions and Constraints

- Users have access to the internet and a supported web browser.
- Users provide correct information.
- The database server remains available during normal operation.
- Payment functionality depends on third-party payment gateway availability.
- The final technology stack may change during implementation.

12. Future Scope

- Skill recommendation based on user interests and activity.
- Ratings and reviews for completed skill exchanges.
- Real-time chat between users.
- Video or audio sessions for online learning.
- Mobile application support.
- Notifications through email or push notifications.
- Advanced search and filtering.
- AI-based skill matching.

13. Conclusion

The Skill Exchange Website provides a structured platform for users to share knowledge and learn new skills from one another. The system supports user registration, profile and skill management, skill discovery, exchange requests, optional payment processing, and administrative management. The requirements described in this SRS provide a foundation for designing, developing, testing, and maintaining the proposed system.

14. References

- Software Requirements Specification (SRS) guidelines and standard software engineering practices.
- UML modeling concepts for software system design.
- Documentation of the selected web, database, and payment technologies used during implementation.